UNSW MARKETING ANALYTICS HACKATHON 2026

Getting slow loans funded faster

What 1.45 million Kiva loans say about predicting slow loans and supporting them

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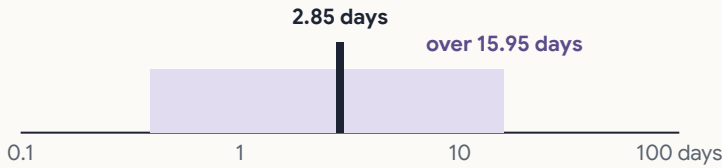


Act on predictable delays now; test writing advice before giving it

Problem, solution and proof across all evidence layers

THE PROBLEM

2.85 days

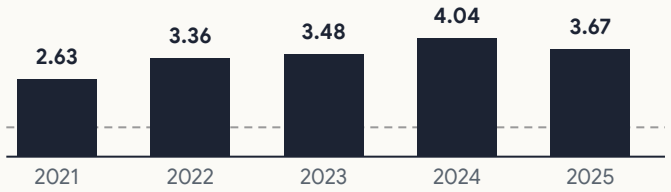


Median wait to full funding

A quarter of loans wait over 15.95 days.

OUR SOLUTION

2.6–4.0×

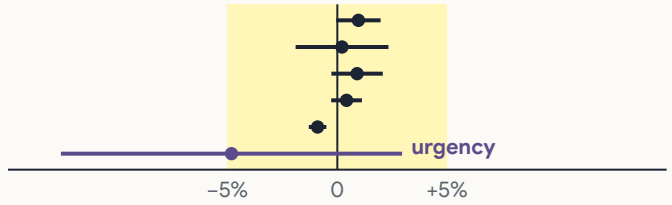


Better than chance at flagging slow loans

Loan details flag the slowest fifth early, so support can go there first.

THE PROOF

±5%



Framing links stay small

Five framings stay within ±5% of funding time, so rewriting stories is not the lever.

Our ask to Kiva: pilot a slow-funding flag with lender-side support, and test one truthful prompt.

Predict Prioritise Test

For whom and when: no period, loan size, gender or region shows a large framing effect; a few rare-word patterns are hypotheses to test.

Agenda

01 → 02 → 03 → 04

The problem

Who waits, and where?

How a loan gets funded 4
The wait 5
Where loans and delays sit 6

What drives the wait

What does the evidence show?

What we read in each loan 7
Key insight 8
Four analyses 9
Country uncertainty 10
Programme controls 11
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For whom and when 13
What text adds 14
Fragile across countries 15

Our solution

What can Kiva do now?

Predict, prioritise, test 16
Predict 17
Prioritise 18
Test 19

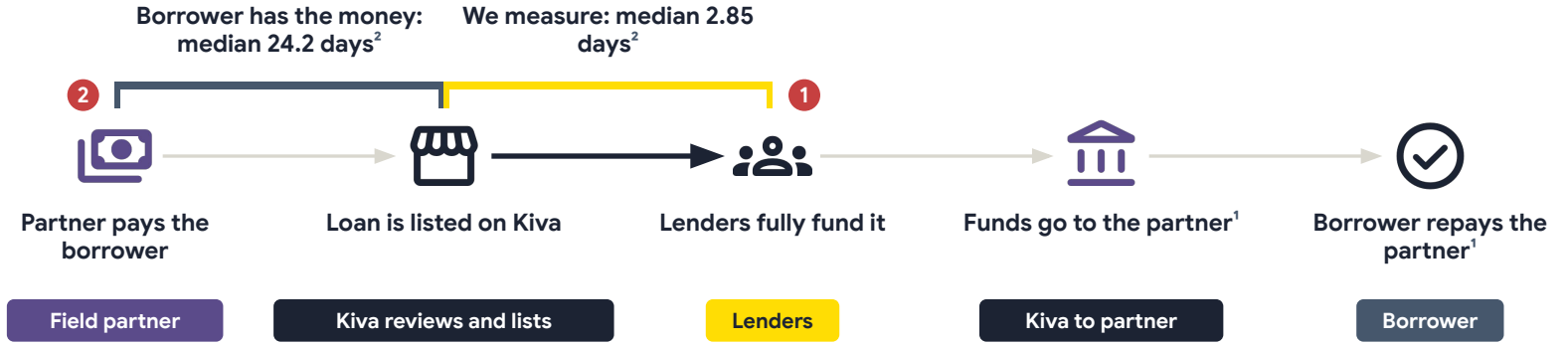
Safe next steps

What keeps it honest, and what comes next?

Guardrails 20
Roadmap 21

Borrowers usually have the money before lenders fund the loan

Descriptive: how a typical pre-disbursed Kiva loan moves, who acts at each stage, and what we measure



Loans disbursed to the borrower before listing



Who is in the data

In the data: 1,453,840 completed loans, including 1,637 refunded

Not in the data: listings that expired or never funded 4

Figure 1. Journey of a typical pre-disbursed Kiva loan, not to scale; medians from the full-data audit.

[1] Typical field-partner flow; these steps are not observed in the data.

[2] Separate medians: 24.2 days covers the 96.4% disbursed before listing, and the two do not add up along one loan.

Scope: 1,453,846 input records; 6 negative durations excluded; 1,637 refunded records included.

1 We study the lender-side step

Our outcome runs from **listing to full funding**, median 2.85 days.

2 Borrowers usually already have the money

96.4% of loans reach the borrower a median **24.2 days** before listing.

3 Speed turns over partner capital

Only **3.6%** are paid out after listing, so slow funding mainly holds up the partner's next loans.

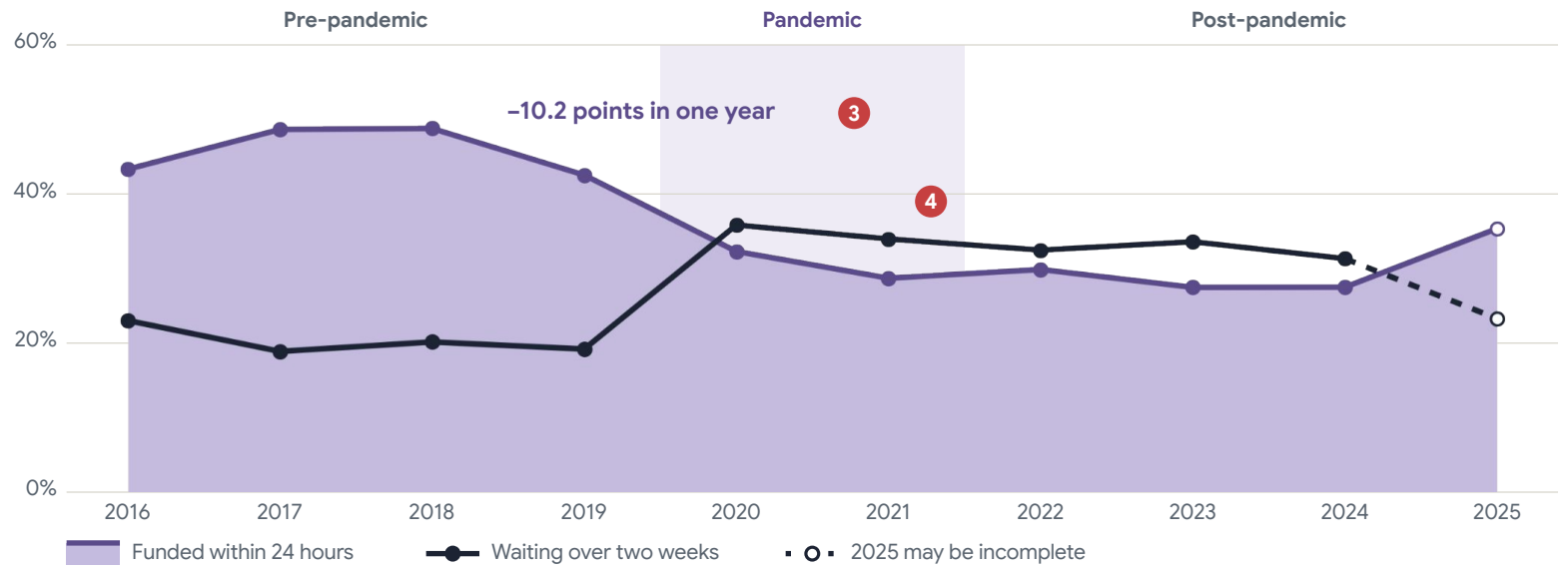
4 Unfunded loans are missing

Only loans that **eventually funded** are in the data; expired listings are not.

→ Why speed is a lender-side problem

Waits got longer after 2019: a third of loans now wait over two weeks

Descriptive: platform funding time from listing to full funding, 1,453,840 completed loans, by posting year



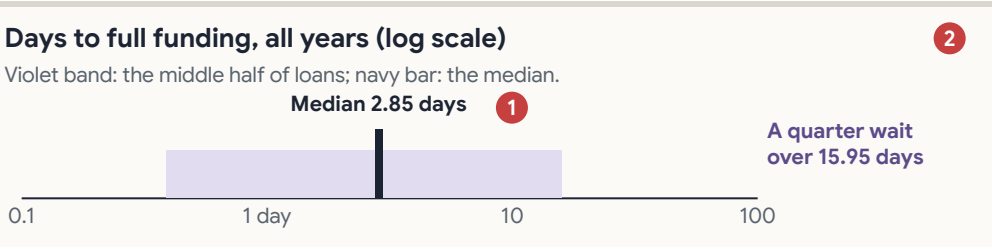




Figure 2. Same-day share and share waiting over two weeks by posting year; funding-time quartiles, all years.

1 Most loans fund within days

Half are fully funded within **2.85 days**, a quarter within **0.39 days**.

2 A long tail waits weeks

A quarter of loans wait **over 15.95 days**, while borrowers already have the money.

3 Same-day funding fell in one year

From **42.5% in 2019 to 32.3% in 2020**, a fall of 10.2 points.

4 Long waits became common

Loans waiting over two weeks rose from **19.2% to 35.8%**; in 2021 that was **55,408 loans**.

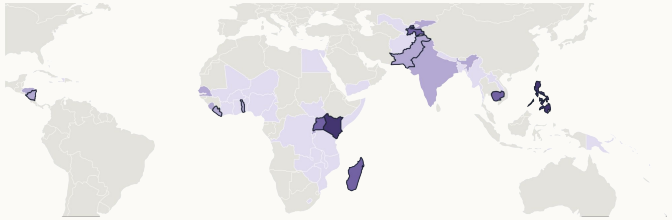
→ Why early warning matters

Over half of all two-week waits come from just three countries

Descriptive: where loans sit and where they wait over two weeks, by country, 1,453,840 completed loans

(a) Where the loans are

Ten of 48 countries hold 81% of loans (outlined on the map); table share cells use the map's colour scale.



	Country	Region	Loans	Share	Same-day
1	**Philippines**	Asia	521,586	35.9%	48%
2	**Kenya**	Africa	266,562	18.3%	29%
3	**Uganda**	Africa	69,651	4.8%	31%
4	**Tajikistan**	Asia	68,865	4.7%	18%
5	**Cambodia**	Asia	58,074	4.0%	49%
6	**Madagascar**	Africa	55,670	3.8%	67%
7	**Pakistan**	Asia	42,767	2.9%	15%
8	**Nicaragua**	Central America	38,520	2.6%	15%
9	**Liberia**	Africa	28,847	2.0%	28%
10	**Togo**	Africa	27,513	1.9%	38%

(b) Where the long waits are

Each bar is all of a country's loans, split by how long they took to fund.

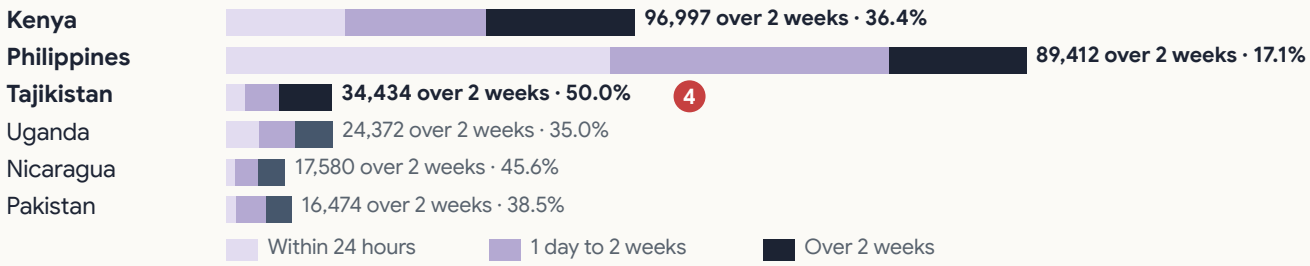


Figure 3. Share of all loans by country (a, map and table) and loans waiting over two weeks (b); same-day: funded within 24 hours.

1 Ten countries hold most loans

Ten of 48 countries hold **81% of loans**; Asia and Africa hold **92.8%**.

2 Speed varies widely by country

Same-day funding runs from **15% in Pakistan to 67% in Madagascar**.

3 Three countries hold half the long waits

Kenya, the Philippines and Tajikistan hold **56.1%** of the **393,847** loans that waited over two weeks.

4 Large markets wait too

In Kenya **36.4%** of loans wait over two weeks; in Tajikistan **50.0%**.

→ Why the flag must work country by country

Each loan story is scored on eight framing measures

Illustrative listing: what is fixed when a loan is listed, and what we measure in the description



FIXED AT LISTING 1

Amount, median	Term, median	Borrowers, median
US$325	**12 months**	**1**
Sector, most common	Country, most common	Partner
Agriculture	**Philippines**	**Field partner**

EDITABLE DESCRIPTION 4

> She has a banana stall at the market to support her family. She wants to buy more stock quickly, before prices rise. She is thankful to every lender who helps her business grow.

The eight framing measures, counted per 100 words, with the share of loans using each (plus a sentiment score) 2

- **Family** — Relatives and schooling — e.g. children · in 88.7%
- **Business** — Trade and stock words — e.g. market · in 85.0%
- **Basic needs** — Water, health, housing — e.g. water · in 23.3%
- **Agency** — Planning and self-direction — e.g. decided · in 54.1%
- **Gratitude** — Thanks and blessings — e.g. thankful · in 19.6%
- **Urgency** — Time-pressure words — e.g. quickly · in 1.3% 3
- **First person** — The borrower's own voice — e.g. my · in 1.0%
- **Third person** — Written about the borrower — e.g. her · in 97.3%

Figure 4. Illustrative composite listing, not a real borrower; values are dataset medians and most common categories.

1 Most of the listing is fixed

Amount, term, sector, country and partner are set; the **description** is the part a partner writes.

2 We score eight framing measures

Eight framing rates per 100 words, plus sentiment and text form.

3 Some measures are rare

Urgency appears in **1.3%** of descriptions and first person in **1.0%**; only **11.3%** never mention family.

4 Pronouns mostly reflect the writer

Descriptions are **written about the borrower**, so third person is context.

→ Why wording is tested, not assumed

Slow funding is a lender-side delay that loan details predict, not wording

Three findings across the outcome audit, the early-warning check and the equivalence test, each leading to a step of the solution

The wait is on the lender side

1

96.4%

of loans reach the borrower before listing



→4

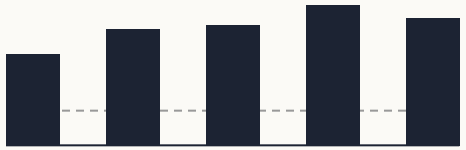
Leads to **Prioritise**

Loan details predict who waits

2

2.6–4.0×

better than chance at flagging the slowest fifth



→17

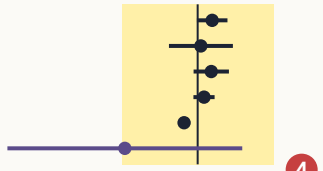
Leads to **Predict**

Wording makes little difference

3

±5%

five framings stay within this share of funding time



→12 →13

Leads to **Test first**

1 Act on the lender side

96.4% of loans reach the borrower before listing, so the wait ties up partner capital, not the borrower.

2 Predict slow loans early

A boosted tree on loan structure flags the slowest fifth **2.6–4.0× better than chance**.

3 Don't chase framing tweaks

Five framings stay **within ±5%** of funding time, and text adds little to the flag.

4 It holds in every group

Six framings stay small in **all 10 subgroups** of period, loan size, gender and region.

→ The evidence behind each insight follows

Four analyses, each answering a different question

One dataset; the equivalence test builds on the robustness checks, and the forecast benchmark asks a separate question

Kiva loans, 2016–2025: 1,453,840 valid completed durations in 48 countries

Explanatory models → 10

Which features are linked to funding speed, and how sure are we?

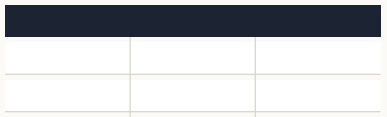
Regressions of funding time and same-day funding; errors clustered by country.

64 of 120 significant terms lost significance

Robustness checks → 11

Do the links survive richer controls?

The same models with more text measures and a programme proxy; multiple-test adjusted.

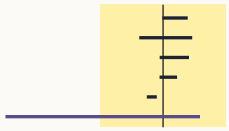


Only small associations remain

Equivalence test → 12

Which links are small enough to rule out a large effect?

Is each 90% interval inside ±5% of funding time?

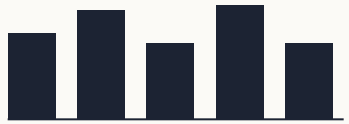


Five framings small; urgency unresolved

Forecast benchmark → 14

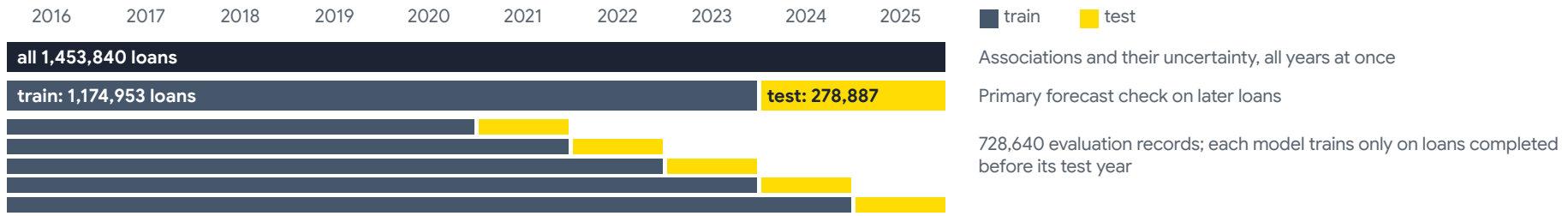
Does text improve a forecast built on loan structure?

Train on earlier years, then test each year with and without text.



Error fell in 5 of 5 years, by 0.26–0.39 days

Years each analysis uses



What each can claim

- **Explanatory:** association
- **Robustness:** survives controls
- **Equivalence:** bounded as small
- **Benchmark:** out-of-time prediction

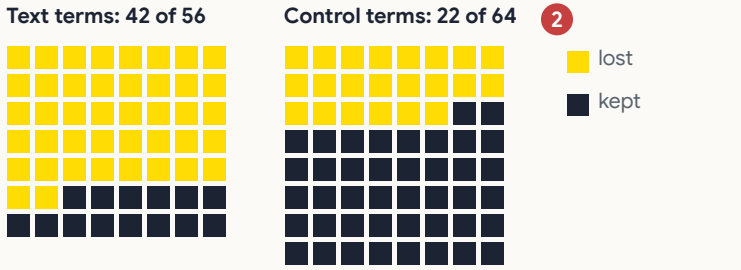
None of these is a causal claim.

Large row counts did not remove country-level uncertainty

Explanatory models: HC3 versus 48-country clustered errors on 1,453,840 loans

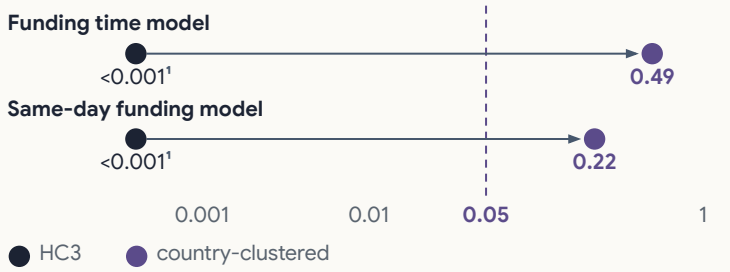
(a) Which terms lost significance

64 of 120 terms significant under HC3 lost it under clustering; each square is one term.



(b) Urgency before and after

HC3 errors allow unequal variance but treat every loan as independent; clustering by country lets loans in a country share shocks.



(c) What it means for the solution

Country-level uncertainty shapes every step of the framework.

1. Report country-clustered uncertainty, never only loan-level errors.
2. Validate the slow-funding flag country by country before using it.
3. Give no country or regional writing rules from two-country regions.

Figure 5. Explanatory models, HC3 versus errors clustered by country (48 clusters).

[1] Printed as 0.0000 in the saved run. Text terms include interactions of a text measure with a control.

1 Text terms are hit hardest

42 of 56 text terms lost significance, against 22 of 64 controls.

2 Few text terms survive

Only **sentiment and some family-framing interactions** with sector or region stay significant.

3 Urgency no longer passes

Its p-value rose from below 0.001 to **0.49** for funding time and 0.22 for same-day funding.

4 Few countries dominate

Ten countries hold **81% of loans**; regional patterns rest on **two countries each**.

→ Why the flag is checked country by country

The only remaining signals are tiny and fail a stricter test

Robustness checks: programme controls, then a wild cluster bootstrap across 48 countries

Term and outcome	Expanded	Programme-adjusted
Third person, log(1 + days)	−0.0090, bootstrap p 0.136	−0.0081, bootstrap p 0.261
Business, 24-hour (pp)	−0.357 pp, bootstrap p 0.337	−0.258 pp, bootstrap p 0.275

Figure 6. Per extra mention per 100 words; BH-adjusted wild cluster bootstrap p across nine terms (Webb weights, 9,999 draws).

0.136 3 1

lowest bootstrap p after BH, any term or model

−0.258 pp 2

business framing, same-day rate per mention, programme-adjusted

1 No signal survives a stricter test

After BH, the lowest **bootstrap p is 0.136**; third person and business no longer pass.

2 What remains is tiny

Business framing is linked to about **−0.258 pp** in same-day funding per extra mention.

3 Pronouns may reflect the writer

Descriptions are usually written by partners, so the **third-person link may reflect who writes**, not a borrower's choice.

→ Why we prioritise loans, not wording

Five framings have small links to funding time; urgency remains unknown

Equivalence test of the robustness-check coefficients; exploratory, bound set after the estimates



Figure 7. Change in funding time per extra mention per 100 words, expanded model, 1,453,840 loans.

[1] Within ±5% of funding time: ±0.05 on log(1 + days), BH-adjusted; bound set after the estimates.

1 Five framings have small links

Family, business, basic needs, agency and third person stay **within ±5%** in both models.

2 Small means under five hours

±5% is about **4.5–4.7 hours** on a typical 2.85-day loan.

3 Urgency is too rare to judge

Under **2% of descriptions** use urgency or first person; their ranges still allow **6–14%** either way.

4 The verdicts hold under stricter tests

A wild cluster bootstrap keeps **35 of 36 verdicts**; three framings stay small even within **±2%**.

→ Why we don't chase framing tweaks

Six framings stay small in every period, loan size, gender and region

Exploratory: programme-adjusted model refitted in 10 subgroups; ±5% equivalence test, bound set after the estimates



Figure 8. Equivalence verdict per framing and subgroup for funding time; programme-adjusted model, 1,453,840 loans.

[1] Unadjusted p-values among about 180 subgroup tests; hypotheses for the Test step, not findings.

1 Small everywhere, not just on average

Six framings stay within ±5% in **all 10 subgroups**; no period or loan-size interaction survives country clustering.

2 Rare words stay unresolved

Urgency is small in only **1 of 10** subgroups and first person in **5 of 10**.

3 Two patterns to test, not to act on

Urgency in Asia and agency by gender stand out among about **180** unadjusted tests.

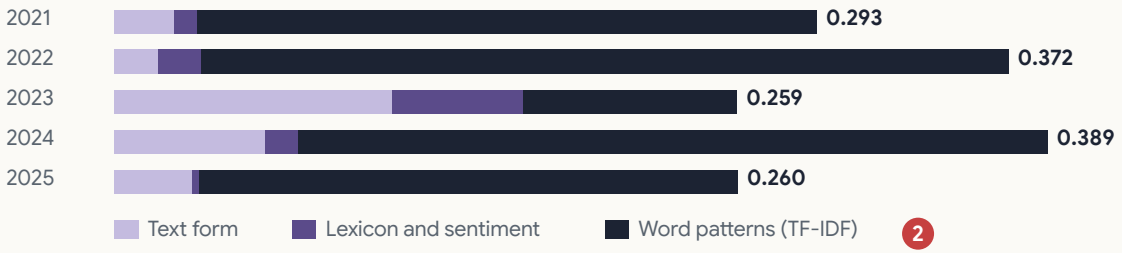
→ Why one approach serves every group

Text adds a little forecast accuracy, mostly from word patterns

Forecast benchmark: linear model gain from adding text, five annual test years, identical rows

(a) Where the gain comes from[1]

MAE gain in days by text feature group, added in order.



(b) Text alone does worse than loan details alone

Extra MAE in days when text replaces loan structure; longer is worse.

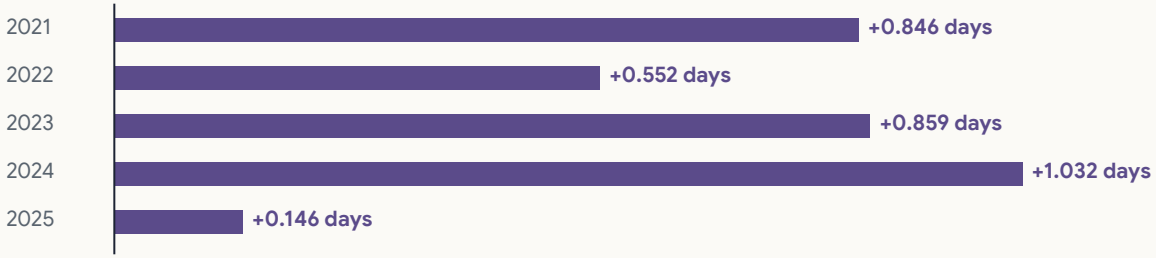


Figure 9. Annual MAE gain in days from adding text (a), and extra error from text alone (b); linear model, identical rows.

[1] Groups added in order: form, then lexicon and sentiment, then TF-IDF; another order would split the gain differently.

1 About seven hours less error

Adding text lowered error in **5 of 5 years**, by 0.26–0.39 days.

2 Word patterns carry most of it

TF-IDF gives the largest share in **4 of 5 years**; lexicon and sentiment add **0.003–0.055 days**.

3 Text works only with loan details

Text alone is worse than loan structure alone by **0.146–1.032 days**, and adds little to the flag.

→ Why we predict with loan details

The text gain holds every year but is fragile across countries

Forecast benchmark: annual MAE gain in days, with 95% ranges from resampling months or countries

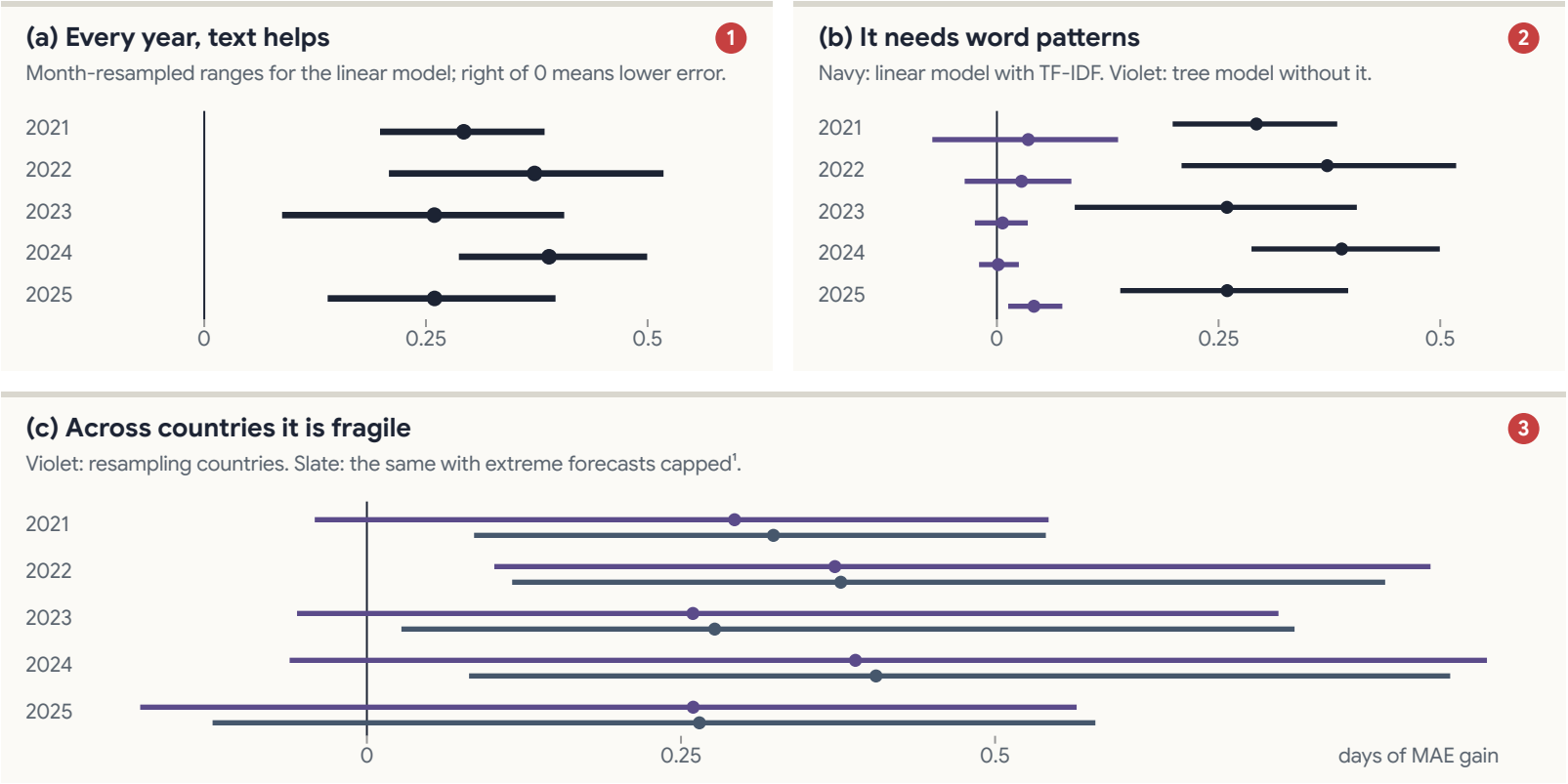


Figure 10. Annual MAE gain in days from adding text, identical evaluation rows.

[1] Forecasts capped at the longest training wait (173.4 days); a diagnostic, not the reported model.

1 Text helps in every year

Month-resampled ranges stay **above zero in 5 of 5 years**; the median gain is about **7 hours**.

2 The gain needs word patterns

The tree model, without TF-IDF, gains **at most 0.04 days**.

3 Across countries it is fragile

Country resampling **includes zero in 4 of 5 years**; capping extreme forecasts leaves zero in **1 of 5**.

→ Why the flag needs validating by country

Our solution: predict slow loans, prioritise support, test before advising

A three-step framework for Kiva and partners, built only on what the evidence supports

01 → 17

Predict

What

Score each listing's risk of slow funding when it is posted.

How

A boosted tree on loan structure; text adds little to this score.

Evidence

Flags the slowest fifth of loans 2.6–4.0× better than chance.

02 → 18

Prioritise

What

Send lender-side support to flagged loans.

How

Featuring, matching funds, lender alerts and partner follow-up.

Evidence

Framing links stay within ±5%, so don't chase framing tweaks.

03 → 19

Test

What

Earn the right to give writing advice.

How

Pass three gates before any advice.

Evidence

No edit to a description has been tested yet.

Writing advice waits until a tested pilot shows it helps.

A slow-funding flag at listing could direct support where it is needed

Concept to pilot: an early-warning card built on the forecast benchmark's retrospective predictions

Listing at posting time

Concept to pilot

Agriculture · US$325 · 12-month term

Illustrative listing, not a real borrower

Predicted risk of slow funding

top 10%

Flag: likely slow to fund

Chance of same-day funding: low

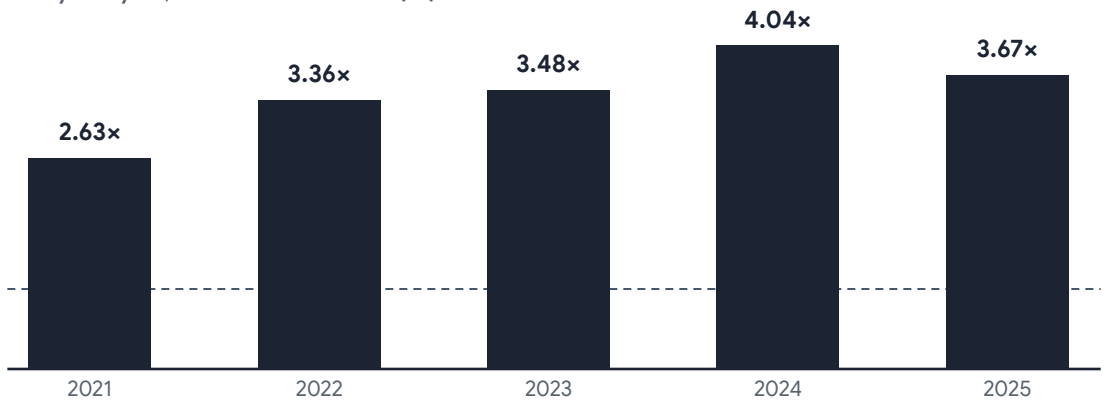
Suggested support

- Feature it to lenders
- Offer matching funds
- Alert lenders who follow the sector
- Ask the partner to review it

2.6–4.0×

better than chance at catching the slowest fifth of loans, flagging the top 10% of predicted-slow loans

Lift by test year; dashed line = random (1×)



Text adds little: the tree's precision changes by −0.003 to +0.005.

Uneven across countries: within the 10 largest, precision ranges 0.18–0.90.

The primary 24-hour model ranks same-day funding with ROC AUC 0.8997 on later loans.

Figure 11. Lift when flagging the top 10% of predicted-slow loans, against the slowest 20% of each test year; boosted tree, loan structure only.

Scope: tested only on loans that eventually funded; a retrospective evaluation, not a deployment test; a flag shows a loan is likely slow, not why. Probabilities need recalibration before use as thresholds.

Send lender-side support to flagged loans, not story edits

Proposed levers for Kiva and partners; their effect on speed is to be tested, not yet measured

Feature it

Show flagged loans to lenders early on the lending page.

Match it

Offer matching funds where a loan is likely to stall.

Alert lenders

Notify lenders who follow the loan's sector or country.

Follow up

Ask the field partner to check the listing is complete and clear.

Don't chase framing tweaks

±5%

Five framings stay within ±5% of funding time, about 4.5–4.7 hours on a typical loan, and text adds little to the flag.

→12





Scope: the levers are proposals; our data do not measure their effect, so each needs a controlled pilot.

Test before advising: three gates before any writing advice

Proposed next research for Kiva and participating partners; each gate must pass before the next

01 → 12

Validate the measures

Question

Do our word counts capture clarity, need and agency?

Do

Build a small codebook; two people code the same narratives and resolve disagreements.

Done when

Definitions are frozen before later periods are checked.

02 → 20

Close the data gaps

Question

What did lenders see, and which listings never funded?

Do

Get original texts, partner identifiers, exposure data and expired listings.

Done when

Unfunded listings and listing-time text are in the data.

03 → 21

Pilot one prompt

Question

Does editing a description change funding speed?

Do

With partners, test one truthful clarity prompt against a control; urgency in Asia is a candidate.

Done when

Threshold, power and guardrails are agreed in advance.

Each gate unlocks a stronger claim

Today
Associations and forecast information

→

After gate 1
Measures that mean what we say

→

After gate 2
Results for every listing, not only funded ones

→

After gate 3
A tested effect of editing

If a gate fails, stop there: report the uncertainty and give no writing advice.

Four guardrails keep the solution honest

What this evidence cannot show, and what to do about it

Associations, not causes

Limit

0 tested edits

Every result is observational; no edit or lever has been tested.

Pilot each lever and prompt against a control. →19

Funded loans only

Limit

Unfunded: missing

Listings that expired unfunded are not in the data, so reach is unknown.

Add expired listings before judging reach. →21

Uneven across countries

Check

0.18–0.90

Flag precision within the 10 largest countries; probabilities drift in 2025.

Recalibrate and validate the flag country by country. →17

For whom and when

Checked

10 of 10

Six framings stay small in every period, loan size, gender and region; rare words stay open.

Test urgency in Asia as a hypothesis. →13

Next: the roadmap from pilot to proof →21

Start with prediction and support now; earn writing advice later

Proposed roadmap for Kiva and partners

Now

1. Pilot the slow-funding flag at listing
2. Route lender-side support to flagged loans
3. Stop investing in framing tweaks

Next

1. Recalibrate and validate the flag by country
2. Validate the text measures with a codebook
3. Pilot one truthful clarity prompt against a control

Later

1. Data partnership: listing-time text, expired listings, partner identifiers
2. Answer for whom and when properly
3. Give writing advice only if the pilot shows it helps

If a step fails, stop there: report the uncertainty and give no writing advice.

Thank you

Mike Nguyen and Sophia Nguyen, Cultural Blend

Code and evidence: github.com/tuannm3812/unsw-ma-hackathon-2026



Regional family links look strong but rest on two countries each

Explanatory models: within-region slopes with country-clustered and heuristic few-cluster intervals



Countries behind each region 3

Africa	27	Central America	2
Asia	12	Middle East	2
Oceania	4	North America	1

Figure A1. Within-region family-framing slopes, 95% ranges, explanatory models; one dot per country, Kiva regions.
[1] t(1) reference on the same clustered error: a heuristic screen, not a calibrated test. Axis in log points × 100.

1 Both links look strong at first

Standard errors put family framing at about **7% faster funding** per extra mention in each region.

2 Two countries cannot carry them

A few-country check widens both ranges **across zero**; the Middle East is **97% Palestine**.

3 Other regions have many more countries

Africa has **27 countries** and Asia **12**; these two regions have two each.

4 No regional writing rules yet

Treat both as **descriptive patterns**; a real test needs more countries or partner-level data.

Each forecast trains only on loans completed before its cutoff

Forecast benchmark: training records must have completed before each cutoff

Evaluation year	Pre-cutoff postings completing later	Eligible as-of training	Evaluation rows
2021	7,528	717,672	163,166
2022	3,918	884,448	143,354
2023	3,546	1,028,174	143,233
2024	5,691[1]	1,169,262	141,725
2025	4,261	1,312,417	137,162

Figure A2. Training availability and evaluation rows by year.

[1] A posting-only split would keep these 5,691 training records; evaluation windows still select eventual completions.

The primary models hold up on later, completed loans

Explanatory models: chronological holdout, completed loans only

5.2031 days

duration MAE, HistGradientBoosting

0.5420

duration R^2

0.8997

24-hour ROC AUC

0.8301

24-hour average precision

0.1156

24-hour Brier score

Scope: 1,174,953 training and 278,887 holdout loans (2024–2025). Retrospective predictions on completed loans, not deployment estimates.

Story themes track loan purpose more than writing style

Exploratory NMF themes from a 20,000-loan sample; descriptive, not full-data evidence

1.5 days

mean funding time, fastest theme (sanitation)

13.5 days

mean funding time, slowest theme (group solar and farm plots)

What the themes show

8 themes, including sanitation, clean water, livestock, smallholder farming and retail. They track loan purpose as much as writing style.

How to read them

Means are unadjusted and come from a sample, so the gap is context, not a tested effect of story content.

Adding text lowered forecast error in 5 of 5 years

Forecast benchmark: linear model, structure only versus structure plus text, identical evaluation rows

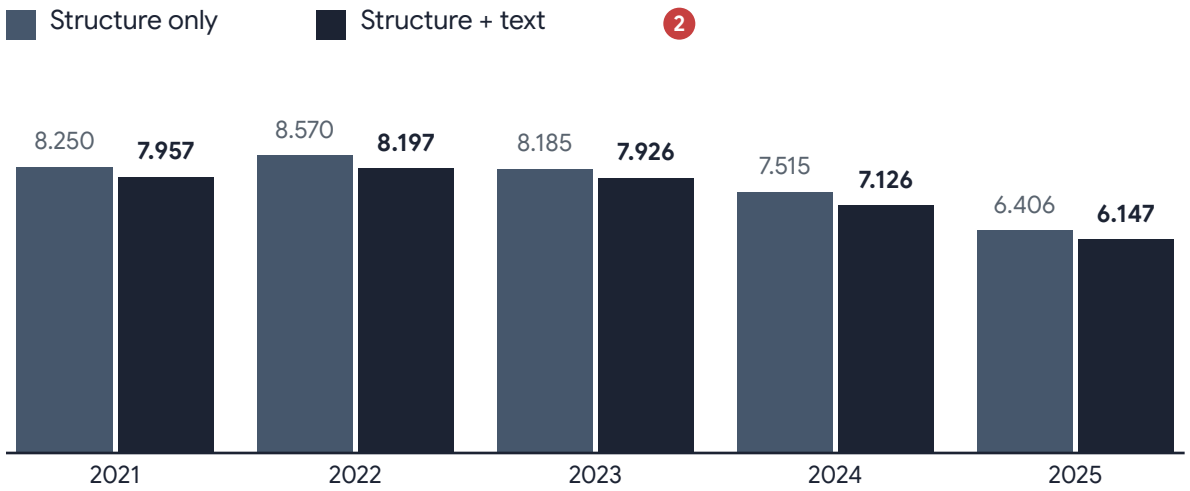


Figure A5. Linear-model MAE in days (top) and its reduction by text feature group (bottom), identical evaluation rows.

[1] Groups added in order: form, then lexicon and sentiment, then TF-IDF; another order would split the gain differently.

0.29 days 1

median annual error reduction

5 of 5 3

years with lower error

728,640

evaluation records

1 Text carries forecast information

Gains of **0.26–0.39 days** on 728,640 evaluation records.

2 Forecasting is not persuasion

Text may encode **purpose, place or programme templates**, not only persuasion.

3 No single country drives it

Removing any single country **leaves the gain positive** in every year (exploratory check).

4 Word patterns carry most of it

TF-IDF gives the largest share in **4 of 5 years**; lexicon and sentiment add only **0.003–0.055 days**.